

AUTOMATIC DRIVING COMPUTING

12. OBJECT TRACKING (1)

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목차

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1. From object detection to object tracking

2. Object tracking – basic principle

3. Hungarian algorithm

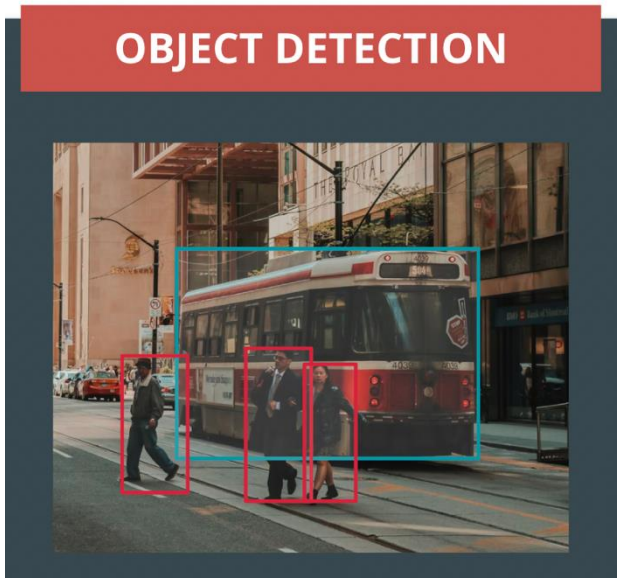
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FROM OBJECT DETECTION TO OBJECT TRACKING

OBJECT DETECTION

Object detection model의 input : single image

output : object별 bounding box 좌표



OBJECT DETECTION

자율주행 시나리오 – input이 single image 아니고 multiple frames

“Bounding box is not the end goal"

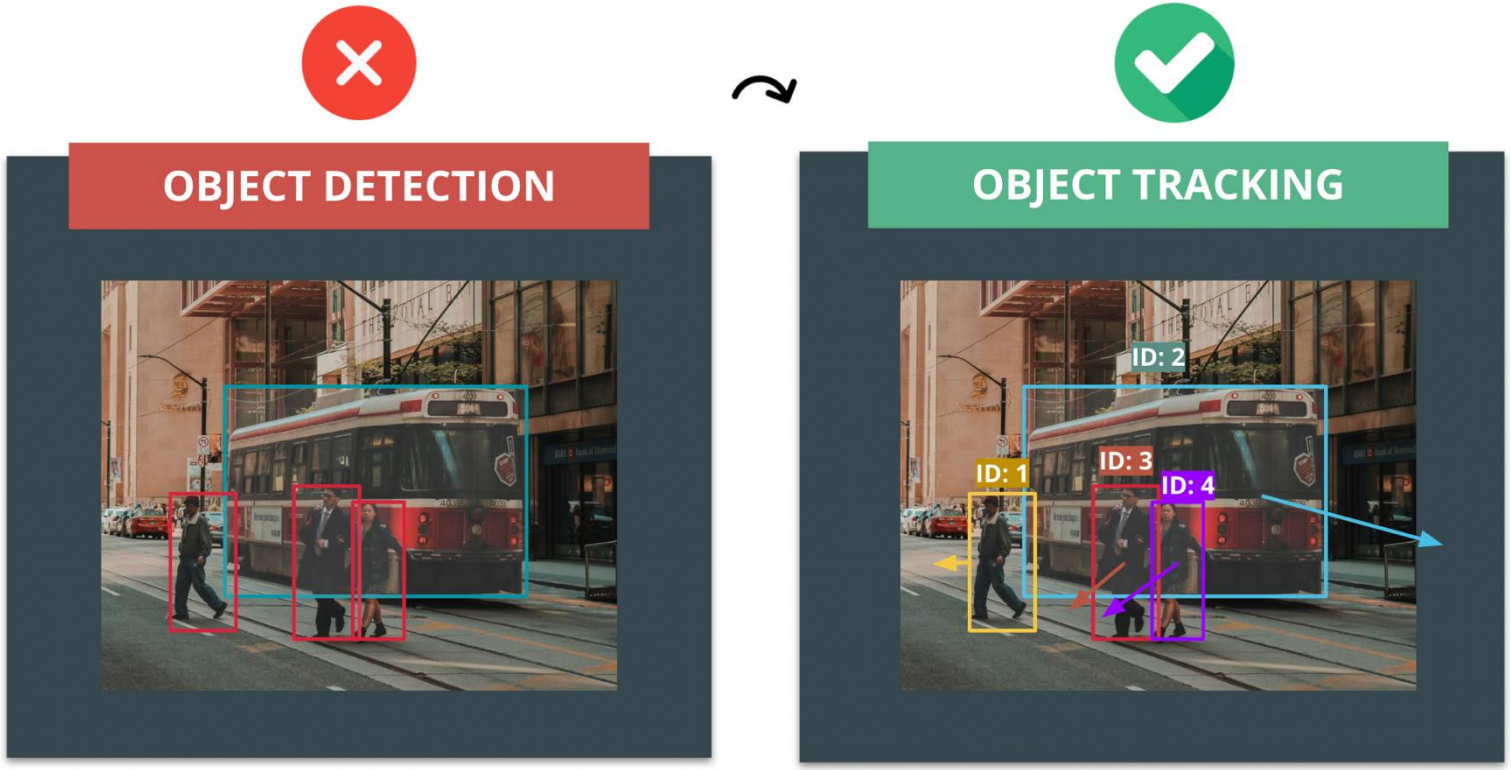


OBJECT TRACKING

Object tracking model의 input : multiple frames

output : object별 bounding box 좌표 + 고유한 ID

➔ 각 ID별 이동 속도 및 방향 등 유용한 정보 취득 가능



OBJECT TRACKING

Example

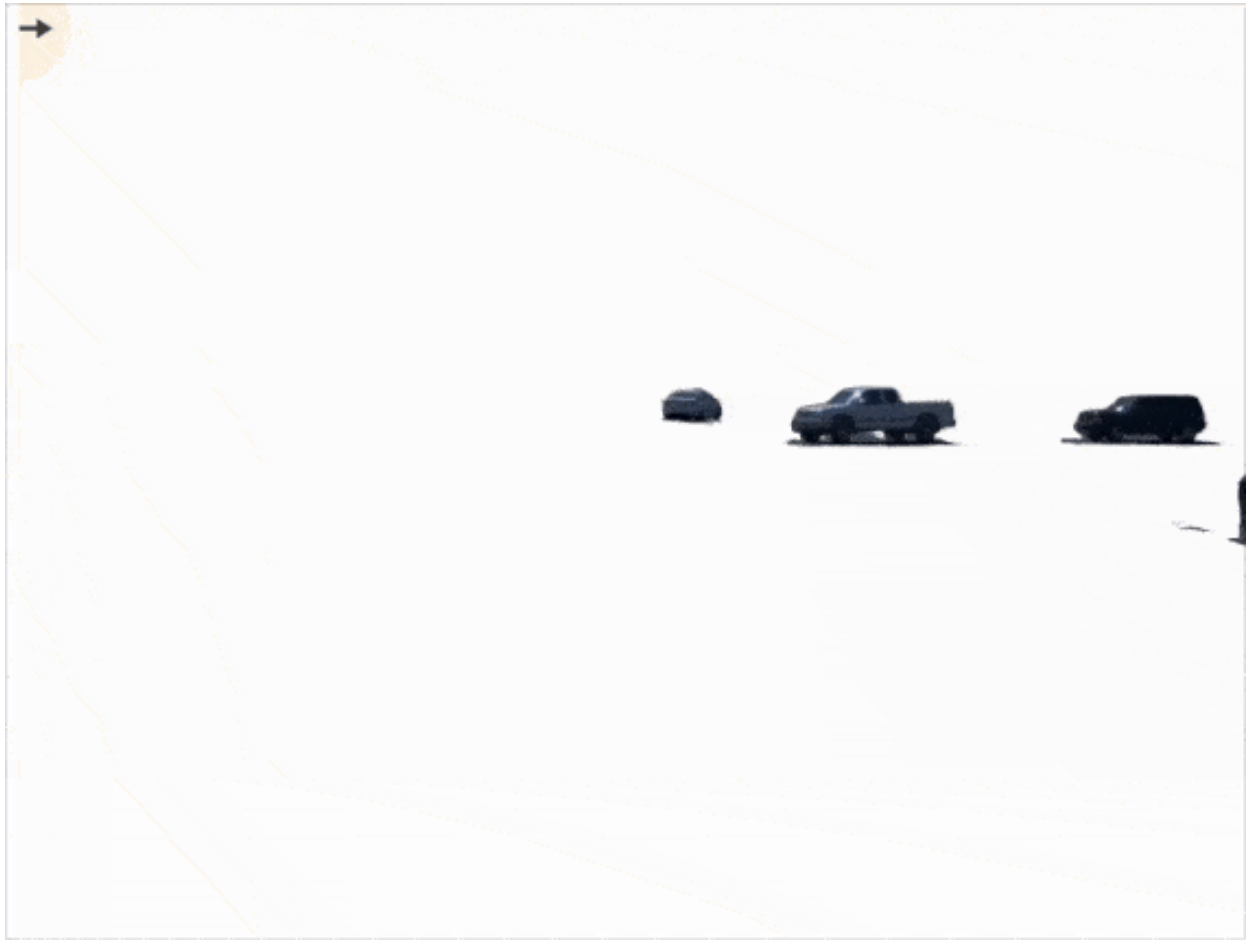
MOT : Multiple Object Tracking



OBJECT TRACKING – BASIC PRINCIPLE

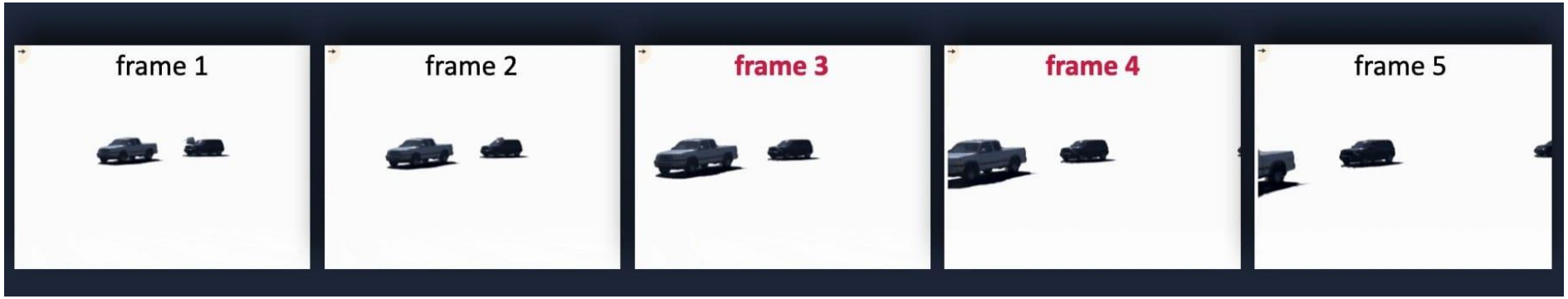
OBJECT TRACKING

Q) How to track (multiple) objects in a video?



OBJECT TRACKING

1) Video를 frame별로 나눔



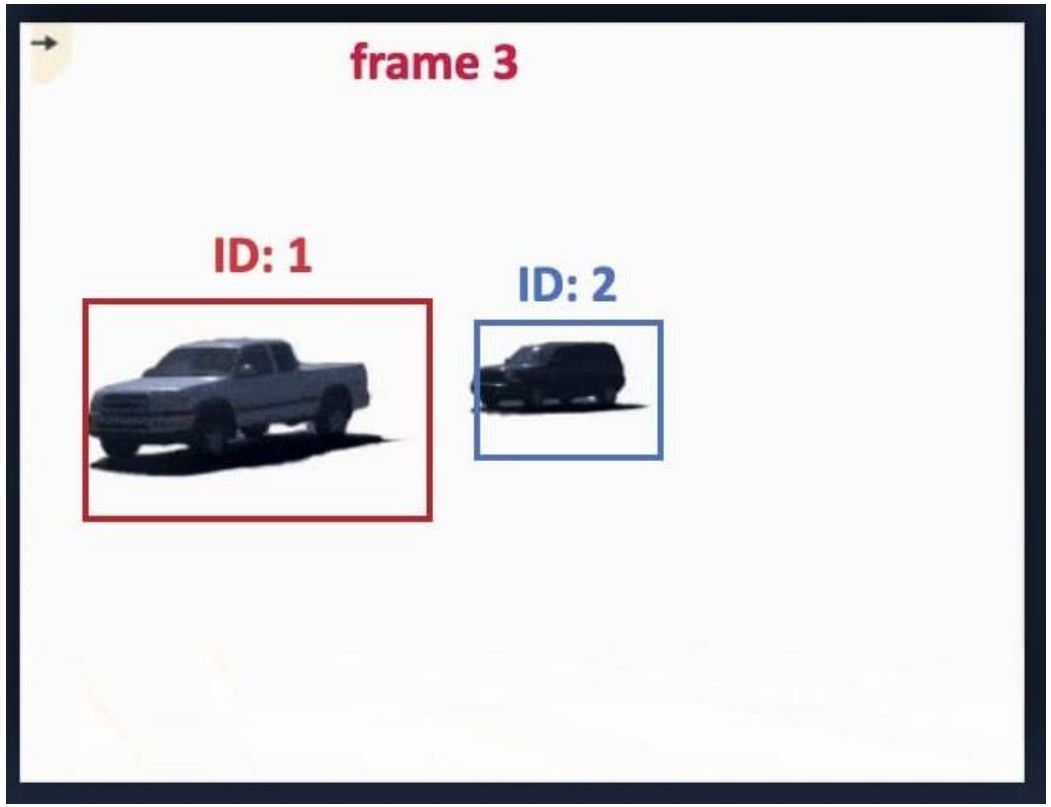
OBJECT TRACKING

2) 각 frame별로 object detector를 돌려서 bounding box를 구함



OBJECT TRACKING

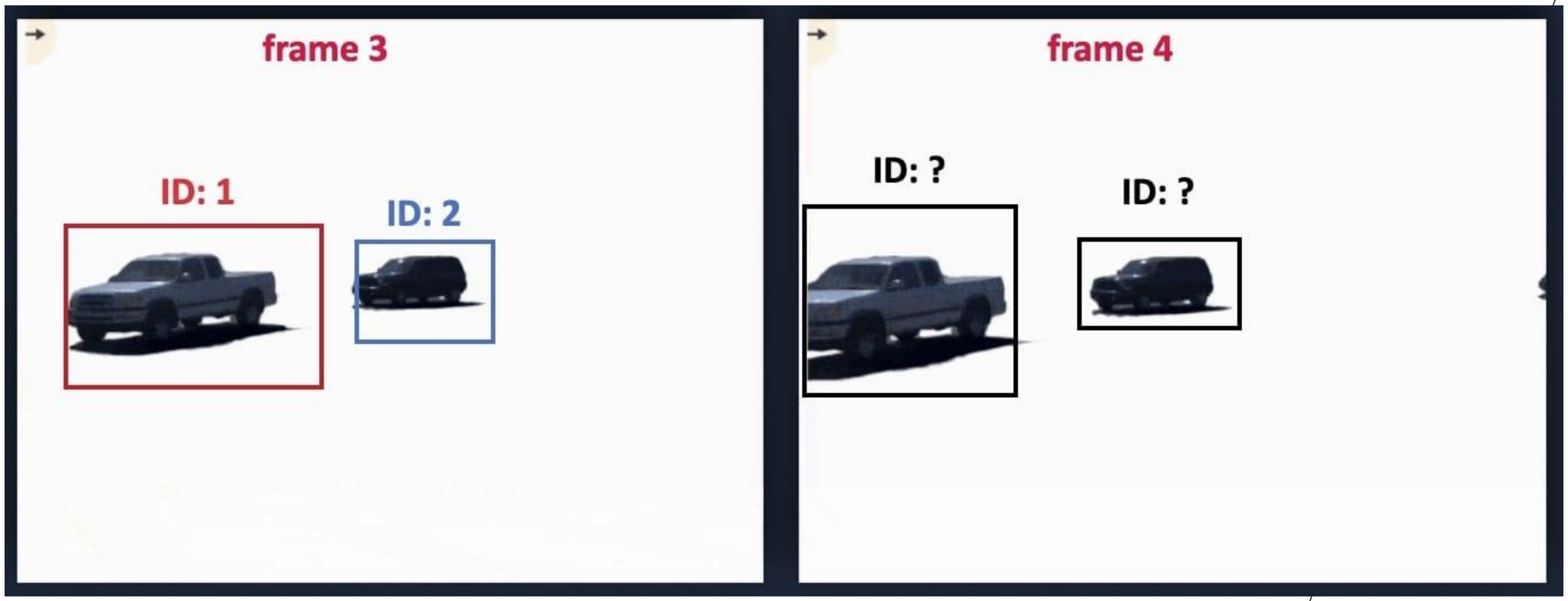
3) 초기 frame 내 각 bounding box마다 initial ID 할당



OBJECT TRACKING

4) 다음 frame 내 bounding box에 ID를 matching

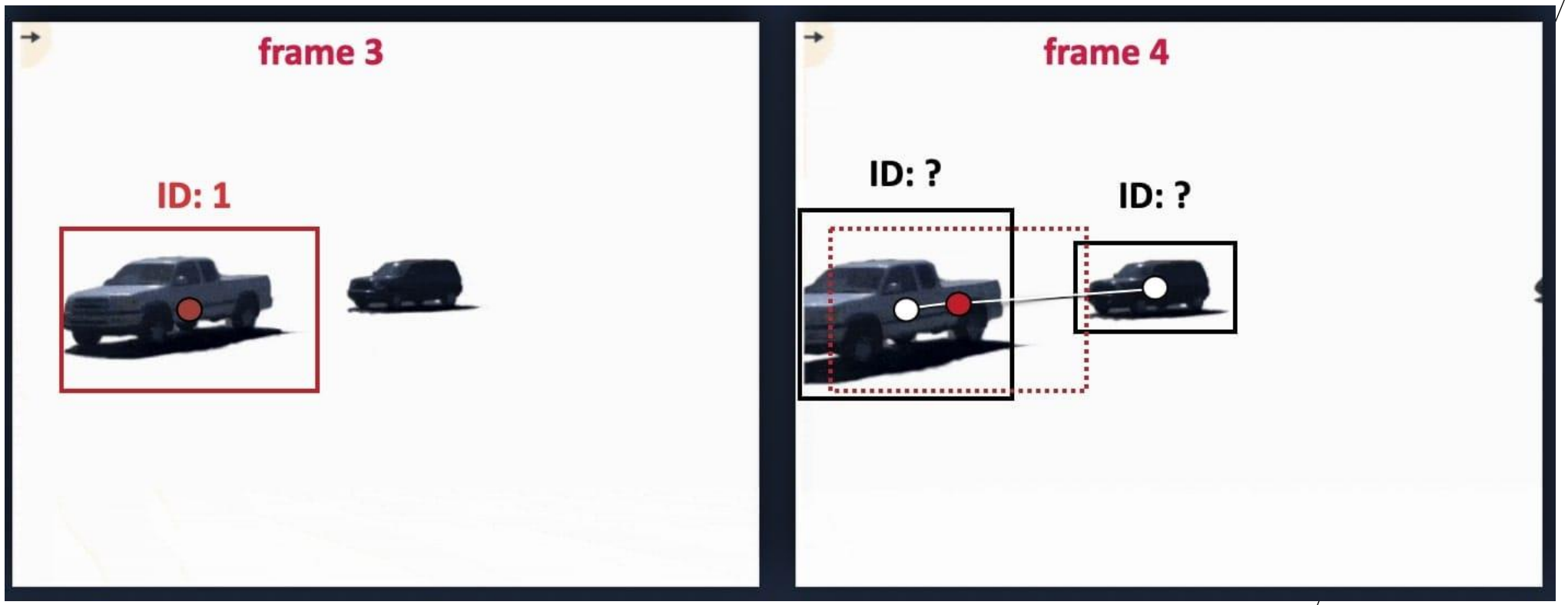
“Data association”



OBJECT TRACKING

4-1) 어떤 “기준”으로 ID matching을 수행?

Ex) bounding box 중심점 사이의 Euclidean distance (= L2 distance)

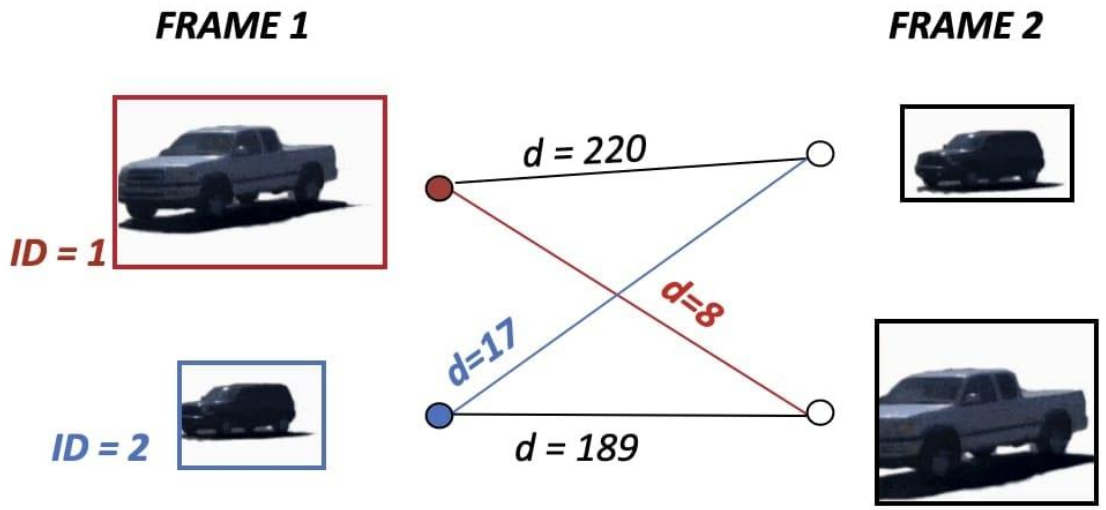


OBJECT TRACKING

4-1)

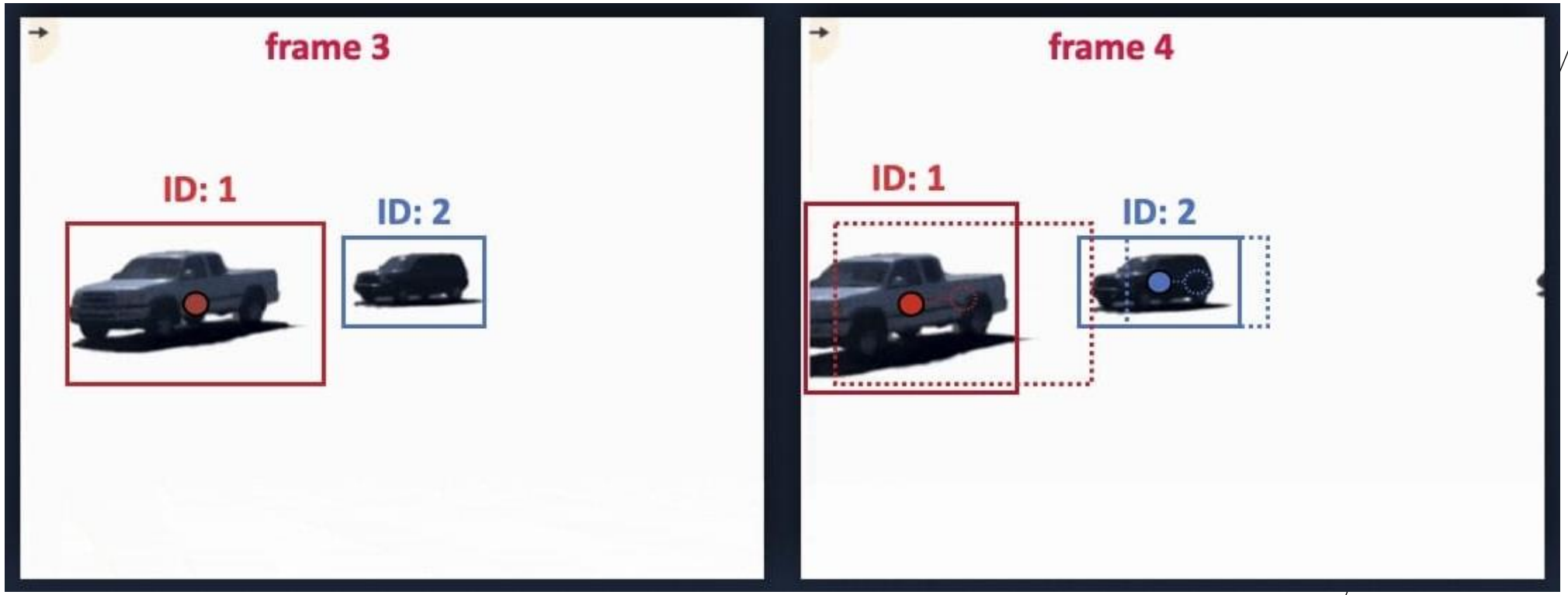
모든 가능한 matching 경우의 수에 대한 distance를 그래프 형태로 표현

➔ Bipartite graph (이분 그래프)



OBJECT TRACKING

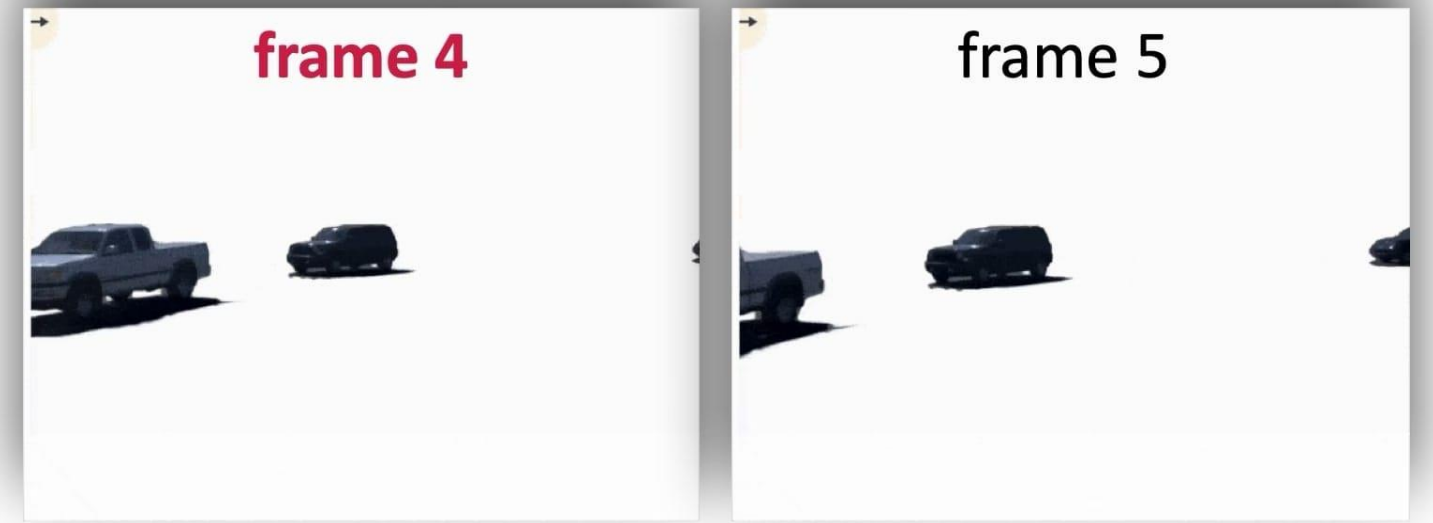
4) Matching result



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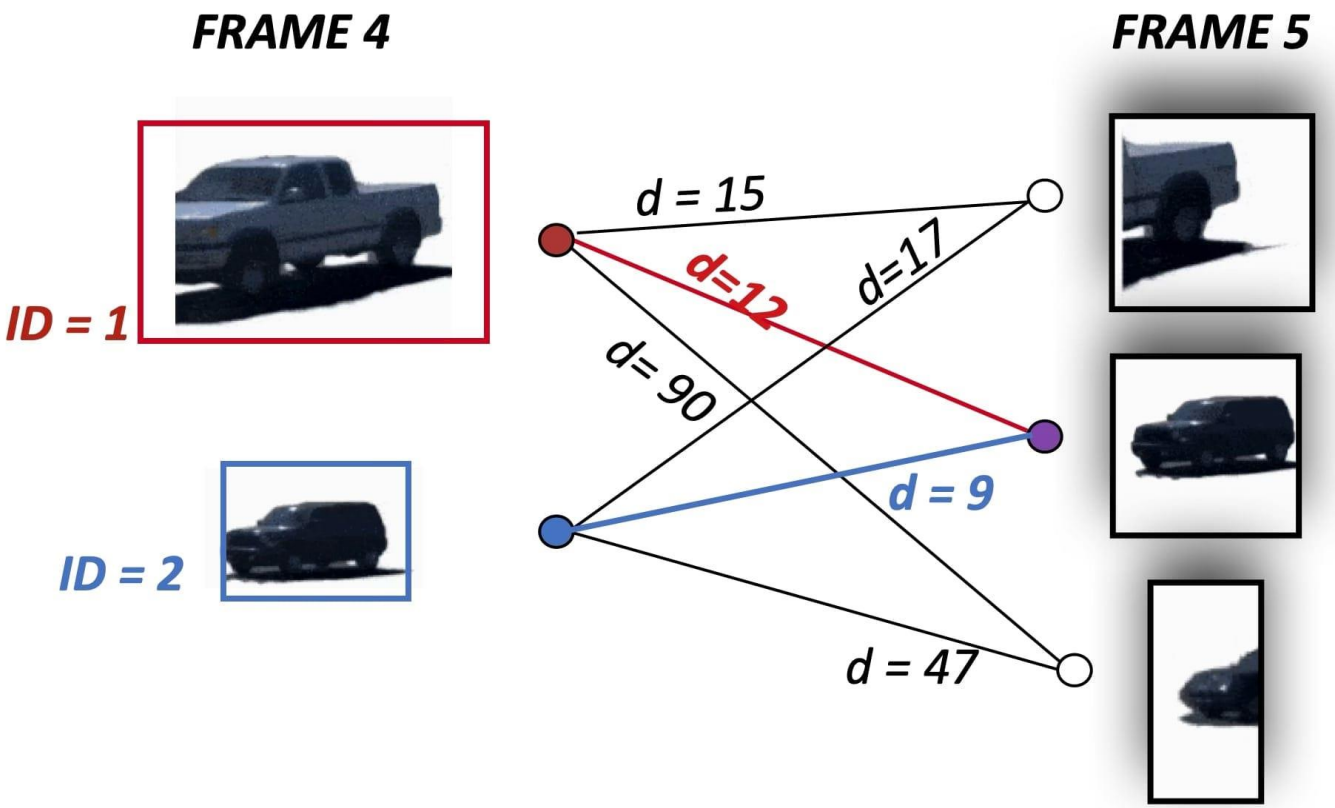
OBJECT TRACKING

Q1) 다음 frame에 새로운 object가 들어온 경우에는?



OBJECT TRACKING

Q2) 최소 거리를 가지는 object가 겹치면?





HUNGARIAN ALGORITHM

HUNGARIAN ALGORITHM

Scenario : 각 Worker에게 Job을 하나씩 할당해줘야 하는 상황

Objective : 일하러 이동하는 거리 합이 최소가 되는 것

	Job 1	Job 2	Job 3
Worker A	10	15	9
Worker B	9	18	5
Worker C	6	14	3

HUNGARIAN ALGORITHM

5 steps of Hungarian algorithm

1) Remove the minimum of each row

	Job 1	Job 2	Job 3
Worker A	10	15	9
Worker B	9	18	5
Worker C	6	14	3



	Job 1	Job 2	Job 3
Worker A	**1**	**6**	**0**
Worker B	**4**	**13**	**0**
Worker C	**3**	**11**	**0**

HUNGARIAN ALGORITHM

5 steps of Hungarian algorithm

2) Remove the minimum of each column

	Job 1	Job 2	Job 3
Worker A	1	6	0
Worker B	4	13	0
Worker C	3	11	0

→

	Job 1	Job 2	Job 3
Worker A	0	0	0
Worker B	3	7	0
Worker C	2	5	0

HUNGARIAN ALGORITHM

5 steps of Hungarian algorithm

3) Cross 0s with the minimum number of lines possible

	Job 1	Job 2	Job 3
Worker A	0	0	0
Worker B	3	7	0
Worker C	2	5	0



	Job 1	Job 2	Job 3
Worker A	0	0	0
Worker B	3	7	0
Worker C	2	5	0



HUNGARIAN ALGORITHM

5 steps of Hungarian algorithm

4) Subtract the smallest element of the matrix. If a number has been crossed twice, add the element to it.

	Job 1	Job 2	Job 3
Worker A	0	0	0
Worker B	3	7	0
Worker C	2	5	0



	Job 1	Job 2	Job 3
Worker A	0	0	2
Worker B	1	5	0
Worker C	0	3	0

HUNGARIAN ALGORITHM

5 steps of Hungarian algorithm

Cross the 0s again and check the number of lines

	Job 1	Job 2	Job 3
Worker A	~~0~~	~~0~~	~~2~~
Worker B	1	5	0
Worker C	0	3	0



	Job 1	Job 2	Job 3
Worker A	~~0~~	~~0~~	~~2~~
Worker B	1	5	0
Worker C	~~0~~	~~3~~	~~0~~

HUNGARIAN ALGORITHM

5 steps of Hungarian algorithm

5) Assign Jobs to Workers starting with the line with only one zero

	Job 1	Job 2	Job 3
Worker A	0	0	2
Worker B	1	5	0
Worker C	0	3	0

HUNGARIAN ALGORITHM

5 steps of Hungarian algorithm

5) Assign Jobs to Workers starting with the line with only one zero

	Job 1	Job 2	Job 3
Worker A	10	**15**	9
Worker B	9	18	**5**
Worker C	**6**	14	3



Job 1	Worker C
Job 2	Worker A
Job 3	Worker B

HUNGARIAN ALGORITHM

In case of NxM, make it NxN by adding columns with the maximum value in each cell

	Tracking 1	Tracking 2	Tracking 3
Detection 1	21	10	0
Detection 2	56	62	11
Detection 3	20	77	90
Detection 4	10	80	20

	Tracking 1	Tracking 2	Tracking 3	Tracking 4
Detection 1	21	10	0	**90**
Detection 2	56	62	11	**90**
Detection 3	20	77	90	**90**
Detection 4	10	80	20	**90**

SUMMARY

SUMMARY

1. **From object detection to object tracking**

2. **Object tracking – basic principle**

3. **Hungarian algorithm**

NEXT LECTURE

Object tracking (2)

QUESTIONS?